

(Git/Github)

EXPERIMENT 26:

Clone an existing GitHub repository to your local machine and make a small change in one of the files.

Tasks:

- Choose an existing public GitHub repository (e.g., your own or a sample repository).
- Clone the repository to your local machine using `git clone <repository_url>`.
- Navigate into the cloned repository directory and open one of the files.
- Make a small edit, such as modifying a line of text or fixing a typo.
- Save your changes.

Aim

To clone an existing GitHub repository to the local machine and make a small modification to one of its files.

Software Requirements

- GitHub Account
- Git
- Visual Studio Code
- Internet Connection

Procedure

Step 1:

Create a Repository (if you don't already have one)

If you already have a repository (for example, **PersonalProject**), you can use that.

Otherwise:

1. Open GitHub.
2. Click + → **New repository**.
3. Name it **PersonalProject**.
4. Select **Public**.
5. Check **Add a README.md**.
6. Click **Create repository**.

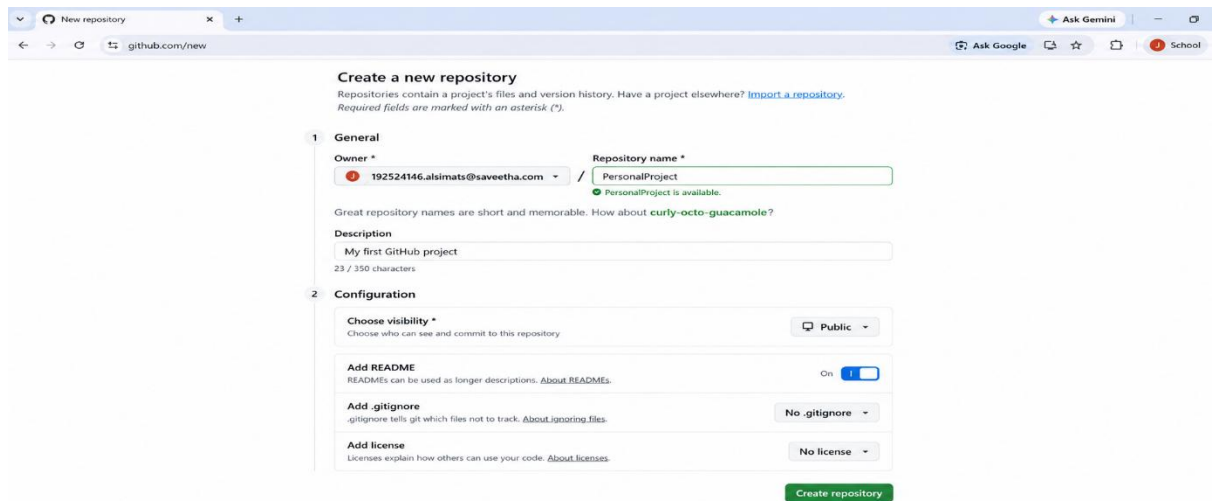


Figure 1: Creation of GitHub Repository

Step 2:

Copy the Repository URL

1. Open the repository.
2. Click the **Code** button.
3. Copy the HTTPS URL.

Example:

`https://github.com/yourusername/PersonalProject.git`

Step 3:

Open VS Code

Open **VS Code**.

Open **Terminal** → **New Terminal**.

Step 4:

Go to Documents

Type:

cd Documents

Press **Enter**.

Step 5:

Clone the Repository

Type:

```
git clone https://github.com/yourusername/PersonalProject.git
```

Press **Enter**.

You should see:

Cloning into 'PersonalProject'...

Receiving objects...

Resolving deltas...

A screenshot of a terminal window with a dark background. The terminal shows the following commands and output:

```
PS C:\Users\yashu\Documents> git --version
git version 2.55.0.windows.3
PS C:\Users\yashu\Documents> git clone https://github.com/yashmitaganesh23-afk/PersonalProject.git
Cloning into 'PersonalProject'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\yashu\Documents> cd PersonalProject
PS C:\Users\yashu\Documents\PersonalProject> code .
PS C:\Users\yashu\Documents\PersonalProject> |
```

 The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. A red error message is visible at the top: '+ FullyQualifiedErrorId : CommandNotFoundException'.

Figure 2: Cloning the Repository

Step 6:

Open the Project Folder

Type:

```
cd PersonalProject
```

Then:

```
code .
```

This opens the project in VS Code.

Step 7:

Open README.md

Click **README.md**.

It may look like:

```
# PersonalProject
```

Step 8:

Make a Small Change

Add one or two lines below it.

Example:

```
# PersonalProject
```

This project is created for the Software Engineering laboratory.

It demonstrates Git repository cloning and file editing.

Press **Ctrl + S**.

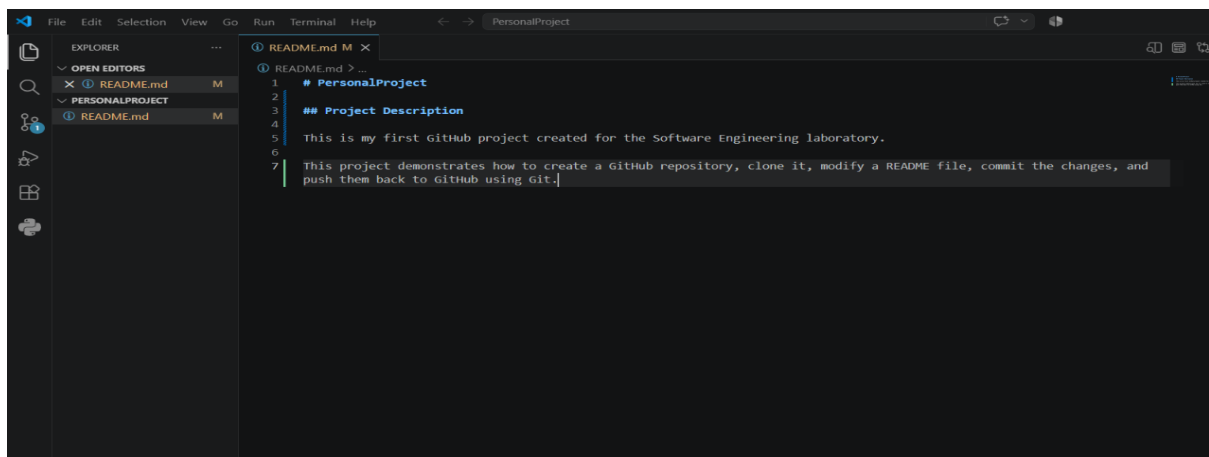


Figure 3: Modified README.md File

Step 9:

Verify the Change

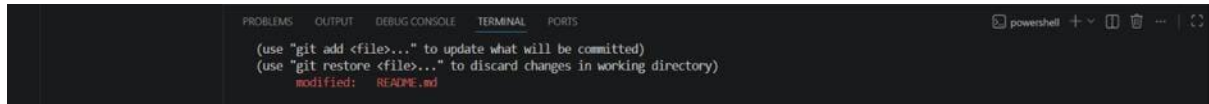
In the terminal, type:

git status

It should display:

modified: README.md

This confirms the file has been modified.



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: README.md
```

Figure 4: Git Status Showing Modified File

Result

Successfully cloned the GitHub repository to the local machine and modified the README.md file.